Evaluating Vision-Language Models in Visual Comprehension for Autonomous Driving

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***Abstract*—Visual cues provide essential information for autonomous driving, while misunderstandings of the visual cues may degrade decision-making quality and put passengers at risk. However, the advent of large language models (LLMs) and vision-language models (VLMs) presents a potential solution to these challenges. These models, trained on vast datasets, demonstrate exceptional capability in understanding context and processing complex information, offering prospects for overcoming perceptual challenges and enhancing autonomous driving systems. This paper investigates the comprehension of visual cues by two large-scale language models, ChatGPT and Bard, through an evaluation involving image-based questions from Tokyo and Beijing driving-license tests. We hope our assessment of the visual capabilities of VLMs in autonomous driving contexts can offer a foundation for future exploration in VLM integration into autonomous driving systems.**

***Index Terms*—Autonomous driving, Vision-language model, Driving-license written test**

I. Introduction

Visual cues play a crucial role in facilitating the safe and orderly movement of vehicles and pedestrians within contemporary traffic systems, offering essential information on routes, directions, and warnings to all participants in traffic [1]. Concurrently, with advancements in autonomous driving technology, there is an increasing imperative for the perception systems of autonomous vehicles to interpret and react to these visual cues with heightened accuracy. This necessity has brought to light certain limitations in the traditional, modular design of autonomous driving systems [2]–[5]. In these systems, distinct components independently handle key tasks such as perception [6], [7], prediction [8]–[10], and planning [11], [12]. Challenges include the potential loss of crucial information during data transfer, the creation of unnecessary computational processes [13], and error accumulation resulting from misaligned optimization goals across modules, potentially impairing decision-making quality [14].

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However, the emergence of large language models (LLMs) and vision language models (VLMs), which have demonstrated proficiency in context comprehension and complex information processing [15], [16], might offer solutions to these challenges in autonomous driving systems. Benefiting from training on expansive datasets, LLMs and VLMs possess a broad common knowledge. Consequently, it is promising to overcome the perceptual network's long-tail challenge and develop more effective autonomous driving systems by harnessing the advanced and extensive capabilities of VLMs for open-world understanding and contextual learning [17]–[20].

To this end, this paper aims to take a preliminary step towards this vision by assessing the comprehension of two large-scale language models, ChatGPT and Bard, regarding visual cues encountered in driving scenarios. Specifically, we compiled 130 image-based questions from the Tokyo driving-license written test and another 130 from Beijing's. The responses of these two VLMs to these questions were then analyzed to evaluate their performance. Due to the absence of a universally acknowledged dataset, our methodology adopts the traffic regulation questions in the driving-license written test, which are accompanied by validated answers, for the purpose of evaluation. We hope our study can provide a foundational evaluation of the visual capabilities of VLMs in the context of autonomous driving, thereby helping to shape the future of VLM integration in autonomous driving systems. The reproduction kit, including the datasets used in the experiments and the experiment results, has been open-sourced and can be found at https://figshare.com/s/1ba5311614041022df47.

The remainder of this paper is structured as follows: Section II provides background information pertinent to this study. Section III outlines the research questions addressed in this study. Section IV details the experimental design and discusses the results obtained. Section V offers conclusion and outlines directions for future research.

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II. Background and Related Work

A. LLMs and VLMs

Large language models (LLMs) have seen significant advancements owing to the proliferation of deep learning in natural language processing (NLP) [21] and the accessibility of extensive public datasets [22]. Presently, LLMs represent a notable development in NLP [23], employing deep learning techniques and voluminous datasets to process, summarize, generate, and forecast new content based on textual data [24]. These models are characterized by their hundreds of billions of parameters, and their training involves massive volumes of text data [25]. LLMs are being actively tested in various domains to assess their applicability in real-world scenarios [26], [27]. Additionally, the 'emergence' phenomenon observed in LLMs has garnered considerable interest. 'Emergence' in this context refers to the appearance of novel behaviors or properties when quantitative changes in a system induce qualitative shifts [28]. This suggests the potential for expanding the scope of LLMs' capabilities.

While LLMs have shown impressive abilities in processing textual data, they traditionally lack the functionality to interpret images. On the other hand, tasks such as visual recognition, image classification, target detection, and semantic segmentation remain central challenges in computer vision research, critical for various applications including autonomous driving. To bridge this gap, Visual Language Models (VLMs) have been developed. VLMs come in various architectures, including the Two-Stream model and the fusion model, but the most advanced implementations often leverage Transformer structures. These structures utilize a self-attention mechanism to process both visual and textual inputs concurrently, efficiently capturing the long-distance dependencies between visual and textual information.

B. LLMs and VLMs in Autonomous Driving

The exploration of multimodal LLMs and VLMs in autonomous driving applications has seen significant advancements to play a crucial role across several autonomous driving functionalities [13], [29], [30]. For instance, in perception tasks, they assist with the tracking and detection of both 2D and 3D objects within driving environments [31]. In the realm of planning and control, LLMs have been leveraged to create motion plans by reformulating these challenges as sequence modeling issues [32]. Additionally, VLMs are instrumental in scene generation, creating detailed and realistic driving videos or scenarios that factor in specific environmental conditions [33].

Despite these innovations, the application of LLM-based methods in autonomous driving encounters notable hurdles. A primary issue is the absence of a standardized evaluation metric, complicating the task of comparing the efficacy of LLM-based strategies against conventional methods. Furthermore, the relatively slow inference speeds associated with LLMs pose significant concerns regarding their practicality for real-time operations within autonomous driving contexts.

III. Research Question

In the process of gathering image questions for the driving-license test, we categorize these questions into three distinct types: (i) those assessing knowledge of traffic signs, (ii) those grounded in particular scenarios, and (iii)those that incorporate elements of both. Consequently, we delineated the following research questions (RQs) to guide our evaluation:

RQ 1: *To What Extent Can VLMs Accurately Interpret Questions Pertaining to Traffic Signs?*

As previously mentioned, traffic signs constitute a fundamental component of contemporary transportation networks, serving essential functions in guiding, directing, and informing road users. Accurate recognition and understanding of these signs are imperative for making informed decisions within traffic contexts. Consequently, assessing VLMs' knowledge of traffic signs is a critical element of the driving-license test process.

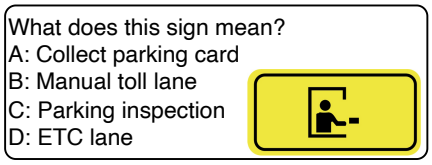

Fig. 1: Example of traffic sign questions.

RQ 2: *To What Extent Can VLMs Accurately Respond to Scenario-Based Questions?*

Scenario-based questions play a pivotal role in driving-license tests. These questions not only test the VLMs' visual comprehension abilities but also examine the VLMs' capacity for envisioning specific driving situations and making appropriate decisions therein. This evaluative mechanism is instrumental in ascertaining an individual's preparedness for the myriad of complex situations they may encounter on the road. Prior research in the literature [34], has delved into ChatGPT's performance on text-based scenario questions within driver's license examinations. Building upon this foundation, our study seeks to extend the inquiry to image-based scenario questions.

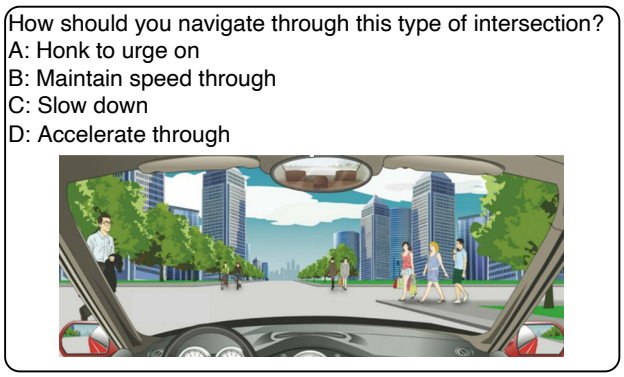

Fig. 2: Example of scenario-based questions.

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RQ 3: *To What Extent Can VLMs Accurately Process Questions Integrating Traffic Signs with Specific Driving Scenarios?*

In real-world driving contexts, traffic signs are often encountered in conjunction with specific scenarios, making questions that combine both elements more representative of actual driving conditions. Such questions, rich in details, not only require attention to various visual cues but also test the ability of individuals to process multifaceted information effectively. This nuanced approach to understanding driving situations presents a unique challenge, one that necessitates a higher order of cognitive processing and decision-making based on a comprehensive analysis of the given information. Given the intricacies of these combined elements, RQ3 seeks to evaluate the accuracy of VLMs in responding to questions that merge traffic signs with specific scenarios.

What should be done when encountering this situation while driving?
A: Change to the middle lane in advance to overtake
B: Change to the far-left lane in advance to overtake
C: Stop behind the school bus and use lights to urge
D: Honk appropriately and overtake from the left at low speed

Fig. 3: Example of scenario questions with traffic signs.

IV. Evaluation

A. Experiment Setting

1) Model of VLMs: The VLMs utilized in our experiment are ChatGPT-4 (version 2023.12) developed by OpenAI, and Bard (version 2023.12, subsequently renamed "Gemini") by Google. Both models were the most current versions available at the time of our experiments.

Our experimental procedure involved manual interactions on the official platforms of ChatGPT and Bard. Due to the restrictions on message volume imposed by OpenAI and Google—specifically, a limit of up to 40 messages every three hours for ChatGPT-4, and a variable limit for Bard based on server demand—repeatedly posing the same question to ensure experimental reproducibility was impractical due to the significant time investment required. Consequently, each question in our study was posed only once.

2) Dataset of questions: In our experiment, we sourced 130 questions related to image-based Japanese written test, along with their answers, from the website " https://menkyoblog.com". This platform offers a comprehensive collection of 26 sets of practice questions for written exams across a variety of topics. Notably, the website is updated with relatively recent test questions, with the newest set corresponding to the 2022 edition. Similarly, we also gathered 130 questions with their respective answers for the Chinese image-based written test from " https://www.jsyks.com". This site is distinguished by its official accreditation and copyright licenses, featuring a test bank that is updated as recently as 2023. Each question is supplemented with an explanation, providing insight into the rationale behind its inclusion. This additional context significantly aids in assessing the logical accuracy of the responses provided by ChatGPT.

However, due to OpenAI and Google's security restrictions, about 10% of the question images could not be processed by Bard or ChatGPT, reducing the actual number of questions evaluated.

3) Prompt design: To ensure the VLMs provided responses to the driving-license written test questions as anticipated, we crafted prompts that positioned the models in distinct roles and scenarios, adhering to a structured format for responding to all questions. Moreover, to eliminate variance in responses attributable to the language of inquiry, we uniformly utilized the official language of our region for the prompts. For instance, when posing questions related to Japanese subjects to ChatGPT, we initiated with a prompt in Japanese: "Imagine you are a Japanese individual taking the Japanese Driver's License Written Exam, and respond to the image-based questions provided. Requirement: state your answer and sequentially explain your reasoning." Furthermore, this experiment framed ChatGPT's responses within the context of zero-shot learning, aiming to evaluate its inherent capability to address questions from the driving-license written test.

B. Experiment Results and Discussion

RQ1: Regarding the image-based questions from the Japanese driving-license written test focusing on traffic sign identification, ChatGPT accurately answered 85 out of 126 questions, achieving an accuracy rate of 67.5%, whereas Bard correctly responded to 84 out of 119 questions, achieving an accuracy rate of 70.6%. In the context of the Chinese questions, both ChatGPT and Bard demonstrated slightly higher accuracy rates within the same category: ChatGPT correctly answered 55 out of 74 questions, achieving an accuracy rate of 74.3%, and Bard correctly answered 54 out of 71 questions, achieving an accuracy rate of 76.1%.

These results indicate that for neither the Japanese nor the Chinese questions did the accuracy rates surpass 80%; notably, the Japanese questions saw accuracy rates not even reaching 71%. This limitation in performance may stem from the absence of specific image information within the training datasets of these VLMs, leading to their inadequate response accuracy. Based on these findings, it is inferred that the VLMs evaluated did not perform optimally when responding to image questions related to traffic signs in the driving-license written test. However, it is important to consider other variables that might influence the experimental outcomes and, by extension, the interpretation of these results. For instance, question difficulty may vary between regions, and the capability of VLMs to process questions in different languages might also differ.

RQ2: In the section pertaining to the Japanese written test, the unique nature of the test content meant that there were

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only a limited number of situational image questions available for evaluation—specifically, just three. Both ChatGPT and Bard managed to correctly answer two out of these three questions, resulting in an accuracy rate of 66.7%. However, given the small sample size and in the interest of maintaining analytical rigor, we have opted not to include these data in our detailed analysis, choosing instead to concentrate on the questions derived from the Chinese section of the test. For the situational image questions from the Chinese written test, ChatGPT demonstrated a higher level of accuracy, correctly answering 78 out of 93 questions, which translates to an accuracy rate of 83.9%. Bard, on the other hand, answered 70 out of 86 questions correctly, achieving an accuracy rate of 81.4%.

This notable improvement in performance when dealing with image questions based on specific situations suggests that the VLMs exhibit a significant capability in this area, with both models achieving accuracy rates exceeding 80%. Such outcomes may be attributed to the emergent phenomenon within the VLMs, which enables them to effectively process and respond to questions grounded in particular scenarios. Despite this improvement and higher accuracy compared to the traffic sign identification questions, the accuracy range of 80% to 85% is still considered insufficient for practical applications, indicating room for further enhancement in the models' capabilities.

RQ3: As mentioned in the discussion of RQ2, the unique content of the Japanese test contributed to a notably small sample size of situational questions, with an absence of questions specifically combining situational questions and traffic sign identification. Consequently, our analysis in this section will solely address the responses to the Chinese questions. In this category, ChatGPT exhibited an accuracy of 81.1%, correctly answering 30 out of 37 questions, while Bard demonstrated a slightly lower accuracy of 80.6%, correctly answering 29 out of 36 questions.

These results reveal that both VLMs managed to achieve over 80% accuracy in answering questions that involved specific situations and also included elements of traffic sign identification. This performance is marginally lower than the accuracy observed for questions that solely focused on situational questions. Despite this, accuracy rates falling between 80% and 85% are considered insufficiently reliable for applications that demand high precision, such as the development of automated driving systems. Nonetheless, the ability of VLMs to process and accurately respond to a variety of detailed information in complex scenarios, maintaining an accuracy rate of 80% or higher, is promising and indicative of the potential capabilities of these models.

V. Conclusion and Future Work

In our study, we evaluated the performance of two VLMs, ChatGPT and Bard, on image-based questions from the driving-license written test in Japan and China. This assessment encompassed 130 image questions from each country, aiming to gauge the models' proficiency in interpreting visual information pertinent to road traffic conditions and to explore their potential applications and limitations in the context of autonomous driving system development.

The aggregate outcomes revealed that ChatGPT achieved a response accuracy of 66.7% for the Japanese image questions and 79.2% for the Chinese ones. Conversely, Bard exhibited a response accuracy of 70.5% for the Japanese questions and 78.5% for the Chinese ones. In addition, the discussion on the research questions indicated that (RQ1) ChatGPT and Bard had low accuracy of about 70% for image questions based on traffic signs; (RQ2) average accuracy of about 80% for image questions based on scenario-based questions; and (RQ3) average accuracy of about 80% for image questions based on traffic signs within a special scenario.

In our future work, we aim to expand our research in the following two directions: Firstly, we intend to introduce other VLMs into our analysis, facilitating a direct comparison of performance variances. Secondly, moving beyond the confines of written-test questions, we plan to develop intricate and lifelike driving scenarios. This approach is designed to provide a more profound insight into the genuine visual understanding and decision-making capabilities of VLMs.

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